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Short communication

Serotonergic and dopaminergic systems are implicated in antidepressant-like effects of *chotosan*, a Kampo formula, in miceSachie Sasaki-Hamada ^a, Azusa Suzuki ^a, Yudai Ueda ^a, Kinzo Matsumoto ^b, Jun-Ichiro Oka ^{a,*}^a Laboratory of Pharmacology, Faculty of Pharmaceutical Sciences, Tokyo University of Science, Japan^b Division of Medicinal Pharmacology, Institute of Natural Medicine, University of Toyama, Japan

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ABSTRACT

We previously demonstrated that *chotosan* (CTS), a traditional herbal formula called *Kampo* medicine, improves diabetes-induced cognitive deficits. In the present study, we investigated the antidepressant-like effects of CTS in mice. The administration of CTS (1.0 g/kg, for 3 days) decreased the immobility time in the forced-swim test, and this decrease was prevented by the prior administration of sulpiride (an antagonist of D_{2/3} receptors) and WAY100635 (an antagonist of 5-HT_{1A} receptors). None of the treatments tested altered the locomotor activity of mice. These results suggest that CTS exerts antidepressant-like effects through changes in the serotonergic and dopaminergic systems.

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Chotosan (CTS) is a Kampo formula that consists of 11 different medical herbs and *gypsum fibrosum*, which is generally prescribed to middle-aged and elderly patients with a weak physical constitution and symptoms related to hypertension and chronic headaches (1). In a clinical study, Terasawa et al. found that the administration of CTS improved impaired activities of daily living, delirium, hallucinations, delusion, and insomnia in patients with vascular dementia (2).

CTS and the anxiolytic drug diazepam were shown to improve anxiety-like behavior in C57BLKS/J-*db/db* mice, an animal model of type 2 diabetes (3), and in senescence-accelerated mice (SAMP8), an animal model of aging (4). Although the neuronal mechanisms underlying the effects of CTS on anxiety-related behavior in SAMP8 and *db/db* mice are not yet clarified, the potential of CTS to improve cognitive and emotional deficits may be beneficial in the treatment of dementia patients.

In the present study, we investigated whether CTS exerts antidepressant-like effects in mice. Monoamine neurotransmitters such as serotonin, noradrenaline and dopamine in the central nervous system are known to play key roles in the pathophysiology of depression (5). We thus also evaluated the involvement of the

serotonergic and dopaminergic systems in the antidepressant-like effects of CTS.

All experimental protocols were approved by the Institutional Animal Care and Use Committee at Tokyo University of Science, and conducted according to the guidelines of the National Institute of Health and the Japanese Pharmacological Society. We used five-week-old male ddY mice (SLC, Shizuoka), and made efforts to minimize the number of animals used, as well as animal pain and distress. All animals were kept in a controlled environment, with a 12:12-h light schedule, temperature of 23 °C, and relative humidity of 55 ± 5% for at least 5 days before experiments were conducted, and were provided *ad libitum* access to food and water.

The CTS extract used in this study was purchased from Tsumura Co. (Japan) in the form of a spray-dried powder extract prepared according to the standardized extraction method of medical plants registered in *Japanese Pharmacopoeia* XV. CTS was extracted from a mixture of 3.0 parts *Uncariae Uncis cum Ramulus* (the hooks and branch of *Uncaria rhynchophylla* MIQUEL), 3.0 parts *Aurantii Nobilis pericarpium* (the skin of *Citrus unshiu* MARKOVICH), 3.0 parts *Pinelliae tuber* (the tuber of *Pinellia ternate* BREITENBACH), 3.0 parts *Ophiopogonis tuber* (the root of *Ophiopogon japonicus* KER-GAWLER), 3.0 parts *Hoelen* (the sclerotium of *Poria cocos* WOLF), 2.0 parts *Ginseng radix* (the root of *Panax ginseng* C.A. MEYER), 2.0 parts *Saposhnikovia radix* (the root and rhizome of *Saposhnikovia divaricata* SCHISCHKIN), 2.0 parts *Chrysanthemi flos* (the flower of

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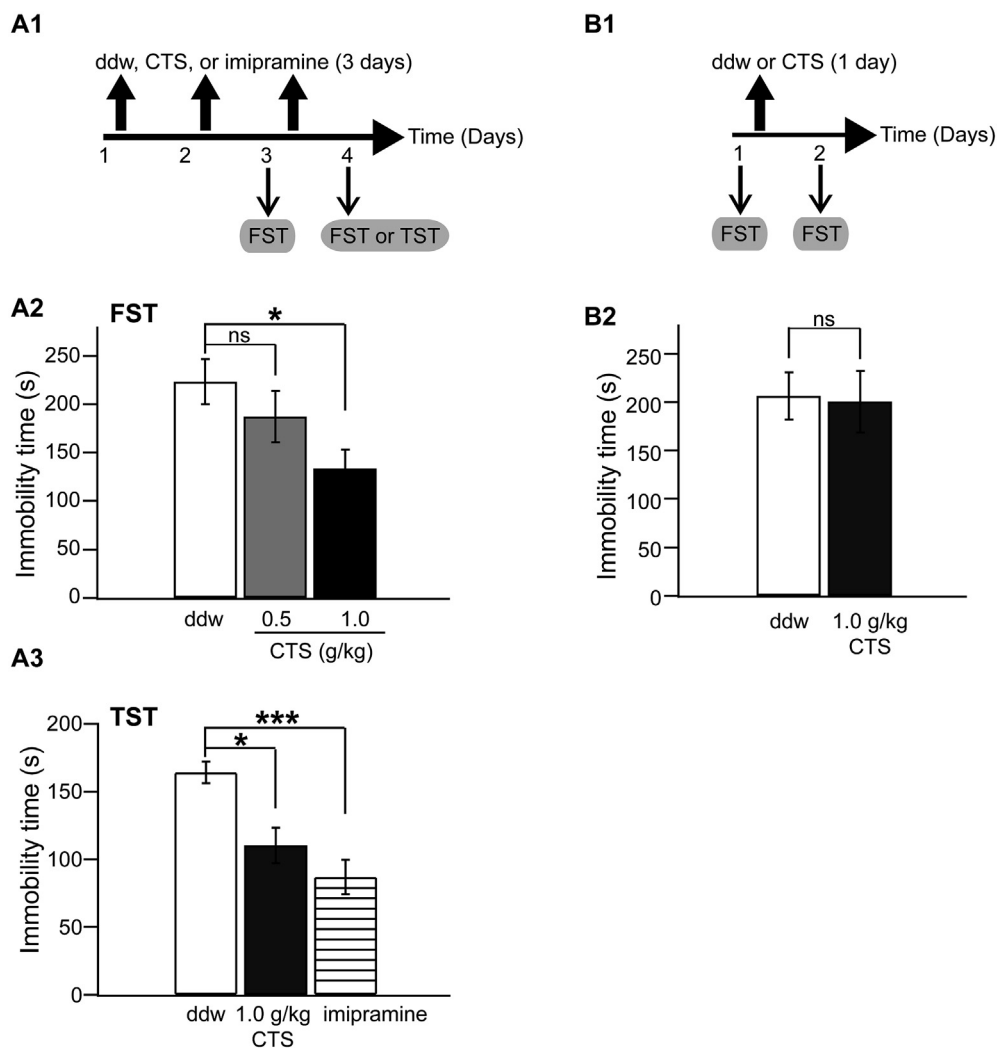


Fig. 1. Effects of CTS on the immobility time in the FST or TST. Experimental schedule (**A1** and **B1**). Effects of CTS (0.5 or 1.0 g/kg/day, p.o., 3 days) or ddw (vehicle) on the immobility time in the FST (**A2**). Effect of CTS (1.0 g/kg/day, p.o., 3 days) on the immobility time in the TST with imipramine as a positive control (**A3**). Effects of CTS (1.0 g/kg/day, p.o., 1 day) or ddw on the immobility time in the FST (**B2**). Results are expressed as means $\pm$ SEM. * $P < 0.05$, *** $P < 0.001$, ns: non-significant [a one-way ANOVA followed by Bonferroni's multiple comparison test (A2 and A3), or Student's t-test (B2)]. $n = 6$, respectively.

Chrysanthemum morifolium RAMATULLE), 1.0 part *Glycyrrhizae radix* (root of *Glycyrrhiza uralensis* FISHER), 1.0 part *Zingiberis rhizoma* (the rhizome of *Zingiber officinale* ROSCOE), and 5.0 parts *Gypsum fibrosum* ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$). The yield of the CTS extract was 16.1%. Protocols for the extraction and chemical profiling of CTS were the same as those described in a previous study (3) and mass spectrometry data obtained from the extract were stored in the Wakan-Yaku Database system (WakanDB ID: LCMS:Chotosan/11000001, <http://wakandb.u-toyama.ac.jp/wiki/LCMS:Chotosan/11000001>), Institute of Natural Medicine, University of Toyama. CTS was dissolved in distilled water (ddw). CTS (1 g/kg) was orally (p.o.) administered at 4–5 p.m. for 3 days according to our previous study (6). All behavioral studies performed at 10–12 a.m.

The forced-swim test (FST) was performed by placing a mouse in an acrylic cylinder (50-cm in height, 18-cm in diameter) containing a 7-cm water column ($25 \pm 1^\circ\text{C}$). Water was replaced between every trial. Two swimming sessions were conducted: an initial 15-min pretest, followed by a 6-min test 24 h later. Test sessions were recorded through a web-camera system in order to measure the time of immobility, with immobility being defined as

floating passively in the water and only making slight movements to keep the head above the water line. The scored immobility time was blindly checked by the co-authors. The following drugs were used: metergoline (Tocris Cookson Ltd., Bristol, U.K.), 4-chloro-DL-phenylalanine (PCPA), imipramine, ketanserin tartrate salt, N-[2-[4-(2-methoxyphenyl)-1-piperazinyl]ethyl]-N-(2-pyridyl)cyclohexanecarboxamide (WAY100635), SCH23390 (Sigma–Aldrich, St. Louis, MO, USA), sulpiride, and yohimbine hydrochloride (Wako Pure Chemical Industries, Osaka). Metergoline (6 mg/kg), SCH23390 (0.03 mg/kg), WAY100635 (0.1 mg/kg), or yohimbine (1 mg/kg) was administered subcutaneously (s.c.), and ketanserin (5 mg/kg), or sulpiride (50 mg/kg) was administered intraperitoneally (i.p.) 45 min before the second swimming session. PCPA (150 mg/kg, i.p.) was pretreated once a day for 4 consecutive days. Six hours after the last PCPA treatment, mice were administered CTS or vehicle. The dosage, time schedules and routes of drug administration were based on previous studies (7,8).

The tail suspension test (TST) was performed. Mice were individually suspended by the tail from a horizontal ring (distance from the floor = 27-cm) in a gray acrylic box ($30 \times 15 \times 15$ cm) using

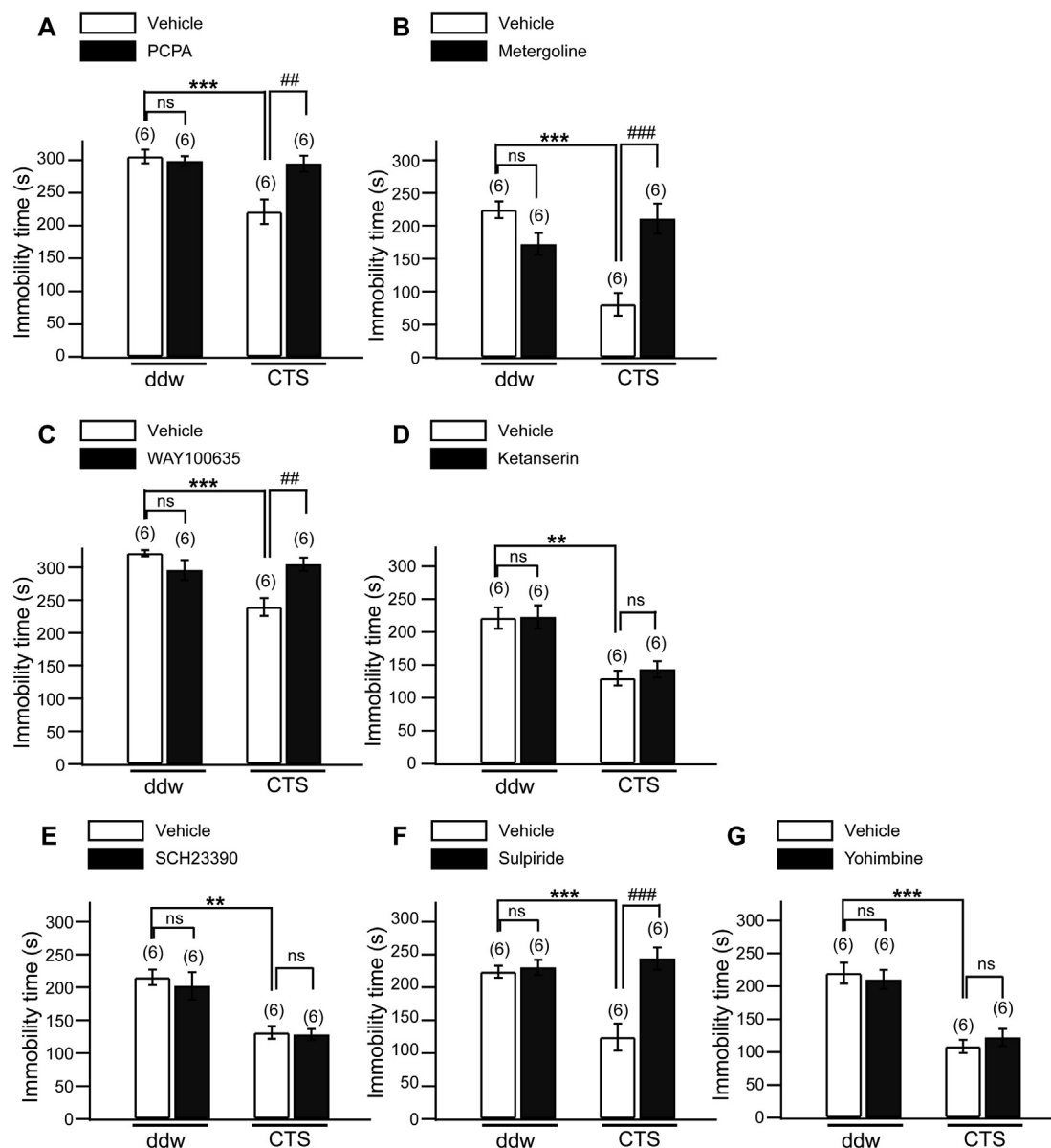


Fig. 2. Effects of the pretreatment with an inhibitor of tryptophan hydroxylase, or antagonists of 5-HT, DA, or α_2 receptors on the antidepressant-like effects of CTS. Effects of the pretreatment with PCPA (150 mg/kg, i.p.) (A), metergoline (6 mg/kg, s.c.) (B), WAY100635 (0.1 mg/kg, s.c.) (C), ketanserin (5 mg/kg, i.p.) (D), SCH23390 (0.03 mg/kg, s.c.) (E), sulpiride (50 mg/kg, i.p.) (F), or yohimbine (1 mg/kg, s.c.) (G) on the antidepressant-like effects of CTS (1.0 g/kg/day, p.o., 3 days). Results are expressed as means $\pm$ SEM. Numbers in parentheses indicate the number of mice used. Comparisons between pharmacological pretreatment and CTS/ddw groups were performed by two-way ANOVA followed by a Bonferroni-adjusted post hoc test. ** $P < 0.01$, *** $P < 0.001$ vs. the ddw group, ## $P < 0.01$, ### $P < 0.001$ vs. the vehicle group, ns: non-significant.

adhesive tape affixed 2-cm from the tip of the tail. A 5-min test session was employed under bright conditions, and was recorded through a web-camera system.

We performed the open-field test (OFT) in another group of mice. The open-field apparatus consisted of a square area (40 $\times$ 40 cm) with 25-cm-high opaque walls. The floor was divided into 16 equal squares by lines. The total number of crossings and the number of crossings in the central area were counted for 5-min. The apparatus was wiped down with paper after the removal of each animal. The significance of differences was evaluated using Student's *t*-test, one-way or two-way ANOVA followed by Bonferroni-adjusted post hoc test. In all cases, significance was set at $P < 0.05$. Statistical analyses were performed using Graphpad Prism (Graphpad Software).

We performed the FST in order to investigate the antidepressant-like effects of CTS. CTS (0.5–1.0 g/kg, p.o. for 3 days) exerted a dose-dependent reduction in the FST immobility time (Fig. 1A2), whereas its single administration did not (Fig. 1B2). Moreover, the oral administration of CTS (1.0 g/kg) or imipramine (30 mg/kg) (a tricyclic antidepressant) once a day for 3 days significantly reduced the TST immobility time (Fig. 1A3).

We investigated the involvement of monoaminergic mechanisms in the CTS (1.0 g/kg, p.o. for 3 days)-induced antidepressant-like effects. Fig. 2A–G shows that the pretreatment with PCPA (an inhibitor of tryptophan hydroxylase), metergoline (an antagonist of non-specific 5-HT receptors), WAY100635 (an antagonist of 5-HT_{1A} receptors), and sulpiride (an antagonist of D_{2/3} receptors) significantly blocked the reduction in FST immobility time elicited by CTS,

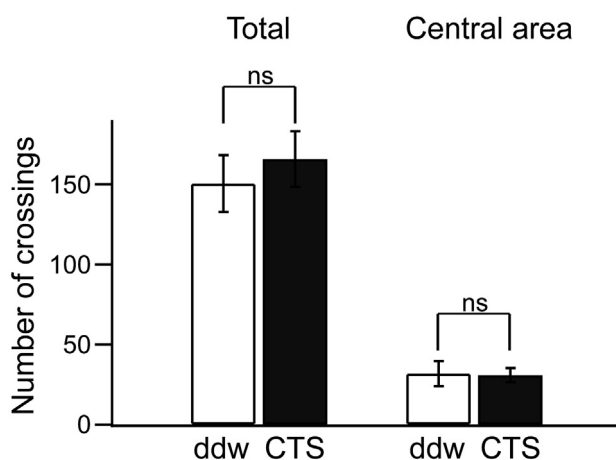


Fig. 3. Effects of CTS on open-field behaviors. The total number of crossings (left) and the number of crossings in the central area (right) were assessed after the administration of CTS (1.0 g/kg/day, p.o., 3 days) or ddw. Results are expressed as means $\pm$ SEM. ns: non-significant (Student's t-test). $n = 6$, respectively.

whereas the pretreatment with, ketanserin (an antagonist of 5-HT_{2A} receptor), SCH23390 (an antagonist of D₁ receptors), and yohimbine (an antagonist of α_2 adrenoceptors) did not.

We also performed the OFT in order to investigate the effects of CTS (1.0 g/kg, p.o. for 3 days) on locomotion activity and emotional responses. CTS had no effect on the total number of crossings and the number of crossings in the central area (Fig. 3).

The antidepressant-like effects of CTS have not yet been examined. We herein showed that CTS reduced the immobility time in the FST without affecting locomotor activity. The FST is widely used screening method for antidepressants in mice. An antidepressant response is characterized by a reduction in the immobility time.

Abnormalities in 5-HT functions have been observed in patients with mood and anxiety disorders (9), and the 5-HT system has been a primary target for drugs that are efficacious in treating these disorders (10). In this study, the pretreatment of metergoline and WAY100635, but not ketanserin inhibited the antidepressant-like effects of CTS. Moreover, PCPA (at a dose known to decrease the content of 5-HT around 60–90% in brain of mice (11)), inhibited the effects of CTS. Based on these results, we suggest that the antidepressant-like effects of CTS involve the release of 5-HT and the activation of 5-HT_{1A} receptors, even though the earlier study reported that the low dose of WAY100635 preferentially blocks presynaptic 5-HT_{1A} receptor mediated responses (12).

The dopaminergic system has also been implicated in the pathophysiology of depression. The results obtained in this study demonstrated that the antidepressant-like effects of CTS may involve the activation of dopamine D_{2/3} receptors. These results suggest that the serotonergic and dopaminergic systems play a role in the antidepressant-like effects of CTS.

In previous *in vitro* studies, geissoschizine methyl ether, which is a plant indole alkaloid and an ingredient of *Chotoko* (*Uncaria cum Ramulus*) contained in CTS, was reported to be a partial agonist of serotonin 5-HT_{1A} receptors (13,14) and a partial agonist or antagonist of dopamine D_{2L} receptors (13). Furthermore, the oral administration of geissoschizine methyl ether ameliorated aggressiveness and reduced sociality (14). We reported that

geissoschizine methyl ether and other alkaloids derived from *Chotoko* were detected in the brain after the oral administration of CTS to mice (15). Future studies are needed in order to clarify whether geissoschizine methyl ether contributes to the antidepressant-like effects of CTS.

In conclusion, we, herein for the first time, showed that CTS exerted antidepressant-like effects in mice. The effects of CTS may involve the participation of 5-HT_{1A} receptors and D₂ receptors. Our results provide further evidence for the beneficial effects of CTS on emotional dysfunctions.

Conflict of interest

There is no conflict of interest to disclose in the present study.

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